CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 5: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A number is divisible by both 5 and 7. It must be divisible by: 1

(a) 40 **(b)** 12 **(c)** 70 **(d)** 35

2. The smallest number which leaves remainder 9 when divided by 16 and 24 is: 1

(a) 57 **(b)** 393 **(c)** 48 **(d)** 105

3. The number of multiples of 25 between 300 and 2000 (both included) is: 1

(a) 71 **(b)** 68 **(c)** 69 **(d)** 70

4. The number of 3-digit numbers divisible by 22 is: 1

(a) 42 **(b)** 41 **(c)** 45 **(d)** 40

5. The smallest 3-digit palindrome divisible by 7 is: 1

(a) 151 **(b)** 168 **(c)** 161 **(d)** 171

6. If the number 3a6b is divisible by 90, then $a + b$ equals: 1

(a) 1 **(b)** 2 **(c)** 0 **(d)** 3

7. The number of steps the Kaprekar routine needs to reach 6174 starting from 1004 is: 1

(a) 9 **(b)** 6 **(c)** 7 **(d)** 8

8. A number is divisible by both 7 and 8. It must be divisible by: 1

(a) 56 **(b)** 15 **(c)** 63 **(d)** 112

9. The smallest number which leaves remainder 11 when divided by 18 and 30 is: 1

(a) 90 **(b)** 551 **(c)** 191 **(d)** 101

10. The number of multiples of 9 between 111 and 999 (both included) is: 1

(a) 100 **(b)** 99 **(c)** 101 **(d)** 98

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The sum of five consecutive numbers is 115. Find the numbers. 2

12. Find the smallest number which leaves remainder 8 when divided by 30 and 45. 2

13. How many multiples of 125 lie between 1000 and 5000 (both included)? 2

14. The sum of five consecutive numbers is 150. Find the numbers. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that the six-digit number 683683 is divisible by 7, 11 and 13. 3

16. Find the greatest 4-digit number divisible by 16, 20 and 30. 3

17. How many 3-digit numbers are divisible by both 12 and 21? 3

OR

Show that the six-digit number 791791 is divisible by 7, 11 and 13. 3

18. Find the greatest 4-digit number divisible by 12, 18 and 27. 3

OR

How many 3-digit numbers are divisible by both 16 and 18? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the digits a and b if the number 3a9b is divisible by 72. List all possibilities. 5

OR

Take the number 865. Subtract its reverse. Add the reverse of the result to the result. What do you get? Show that this always happens for 3-digit numbers whose first and last digits differ by 2 or more. 5

20. Find the digits a and b if the number 2a5b is divisible by 18. List all possibilities. 5

OR

Take the number 971. Subtract its reverse. Add the reverse of the result to the result. What do you get? Show that this always happens for 3-digit numbers whose first and last digits differ by 2 or more. 5

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND